

Project: Development of Vantare VantageSuit™: A Real-Time Biomechanical Feedback Exosuit

Start Date: 1st June, 2026

Duration: 10 Weeks

Priority: Top

Client: Mark

Dear Muhammad,

Thank you for following up with me and I also appreciate your patience, as I was visiting a location with limited internet access.

I have now had the opportunity to carefully review the proposal in detail. Overall, I think the proposal is very strong and comprehensive, and I appreciate the level of thought and structure that went into it.

Before moving forward, I would just like to clarify a few important items to ensure we are fully aligned on deliverables and expectations for this phase of the project:

I would like confirmation that the final deliverable will be a fully working integrated prototype system, including the wearable suit, sensors, PCB system, firmware, BLE communication, haptic feedback integration, and mobile application functionality, rather than only research, architecture, or partial demonstrations.

Can you please clarify exactly what is meant by “3 prototypes included”? I would like to better understand whether this refers to:

- 3 complete wearable prototype systems
- 3 PCB iterations
- 3 hardware revisions
- or another interpretation

Please confirm exactly what is included within the \$14,000 package pricing, specifically regarding:

- components
- PCB fabrication
- suit materials
- assembly
- firmware development
- Flutter mobile app development
- testing
- shipping/delivery
- and any third-party software or manufacturing costs

I would also like confirmation that all final ownership and source materials will be transferred to me upon completion, including:

- CAD files
- STEP files
- PCB files
- Gerber files
- BOM
- firmware source code
- Flutter app source code
- credentials/accounts
- technical documentation
- and related project assets

Regarding the mobile application, I would like confirmation that the delivered app will include functional implementation of:

- BLE pairing/connectivity
- live movement monitoring
- session tracking
- coach workflows
- export functionality
- and tested Android/iOS builds

rather than only UI/UX mockups or architecture planning.

Finally, I would like clarification that real-time haptic feedback functionality is included within the delivered working prototype for this phase, rather than being reserved solely for future expansion phases.

Additionally, when possible, I would likely have the current engineering team working on Phase I send the current exosuit and related materials directly to your team for integration and continuation work. Please send over the preferred shipping address and recipient information for that process when convenient.

I appreciate all of the work and effort that has gone into this proposal and look forward to your clarification on the above points so we can ensure everything is fully aligned moving forward.

Best regards,
Mark Worrell

Dear Mr. Mark,

Thank you for your detailed review and thoughtful questions. Please find the clarifications below:

1. Yes, upon project completion, full ownership and transfer of all project assets, source files, documentation, and related materials will be provided to you.

2. The “3 prototypes included” refers to 3 fully fabricated and complete wearable prototype systems, including replacement warranty support if required during the development phase.

3. package includes all major project-related costs, including components, PCB fabrication, suit materials, assembly, firmware development, Flutter mobile application development, testing, and complete system integration.

4. Yes, the delivered system will include functional BLE communication, live monitoring capabilities, mobile application functionality, and all related core features discussed in the proposal.

5. Yes, real-time haptic feedback integration and functionality are included within the delivered prototype system for this phase.

6. The preferred shipping address and recipient details for sending the current exosuit and related materials will be shared once the Upwork contract is officially initiated.

I appreciate your thoroughness and professionalism throughout this discussion, and I look forward to moving ahead together on this project.

Best regards,
Muhammad Bilal

Hi Muhammad,

I also wanted to provide one additional clarification regarding both my understanding of the current Phase II scope and the long-term final product vision for the Vantare VantageSuit™ platform so we are fully aligned moving forward.

For the current Phase II development scope, my understanding is that the objective is to deliver a functioning integrated prototype platform that includes:

- real-time biomechanical movement monitoring
- wearable sensor integration
- BLE communication between hardware and mobile application
- session start/stop capability
- movement data collection and retrieval
- session summaries and performance visualization
- functional Android/iOS mobile application
- real-time monitoring interface
- foundational movement comparison and deviation detection workflows
- foundational haptic feedback triggering capability
- coach-defined workflow and movement configuration foundation
- export functionality including CSV/JSON/raw data where applicable
- and integrated hardware, firmware, PCB, and application-level communication

Additionally, I want to clarify my vision for the final evolved Vantare VantageSuit™ platform beyond the current phase.

For the final intended product vision, the long-term objective is for coaches and/or AI systems to be able to program movement guidance workflows within the application that help guide exosuit users toward optimal movement patterns in real time. The system would monitor user movement, collect biomechanical data, provide overall session summaries, and identify movement deviations from coach-defined or AI-defined standards.

When the user moves away from the desired movement pattern, the exosuit would provide real-time haptic/vibration-based corrective feedback intended to guide the user back toward the recommended movement pattern, ultimately supporting motor learning, movement reinforcement, skill acquisition, and performance development workflows.

My understanding is that the current Phase II scope establishes the foundational intelligent prototype architecture and core systems necessary to support future evolution toward this long-term platform vision.

I simply wanted to ensure alignment regarding both the current prototype scope and the intended long-term direction of the platform moving forward.

Thank you again, and I look forward to your feedback.

Best regards,
Mark Worrell

Thank you for the additional clarification and for outlining both the current Phase II scope and the long-term vision for the Vantare VantageSuit™ platform.

Yes, we are fully aligned with your understanding of the Phase II objectives. The goal of this phase is to deliver a functioning integrated prototype platform including wearable hardware, sensor integration, PCB systems, firmware, BLE communication, mobile

application functionality, movement monitoring, session management, foundational deviation detection workflows, haptic feedback capability, and complete hardware/software communication.

We also fully understand the broader long-term vision for the platform, including AI-assisted and coach-defined movement guidance workflows with real-time corrective haptic feedback, biomechanical analysis, and performance development capabilities.

This phase is being developed with a scalable and modular architecture to support both the immediate prototype goals and future platform expansion.

Thank you again for the detailed clarification. We are excited to move forward together on this project.

Best regards,
Muhammad Bilal
Let's get started Mr. Mark!